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Piperacillin/tazobactum and cefotaxime decrease the effect of beta lactamase production in multi-drug resistant *K. pneumoniae*

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ABSTRACT

Background: Worldwide, multi-drug resistant *Klebsiella pneumoniae* (*K. pneumonia*) and their virulence's were contributed more in the multi-drug resistant effect. According to the World Health organization report, it is an emerging thread in developing countries and also comes under first ever critical list. In this context, the current study was concentrated on detection of extended spectrum beta lactamase (ESBL) producing strain and their antimicrobial susceptibility study of *K. pneumoniae*.
Materials and methods: Firstly, the multi-drug resistant effect of the *K. pneumoniae* was identified from specific CLSI guidelines recommended antibiotics by disc diffusion method. Consecutively, the primary ESBL identification test was performed using ceftazidime and cefotaxime, followed by double disc combination and phenotypic confirmation tests using ceftazidime/clavulanic acid and cefotaxime/clavulanic acid. Finally, the minimum inhibition concentration of some important sensitive antibiotics were performed against selected *K. pneumoniae* was confirmed by micro broth dilution method.
Results and conclusions: The current result was most favorable to selected *K. pneumoniae* with more multi drug resistant characteristic nature. All the performed antibiotics were almost more sensitive to selected *K. pneumoniae*. The effective antibiotics of piperacillin was also exhibited more resistant effect against tested bacteria and it cleaved the bacterial enzyme clearly. The present result of primary ESBL identification test result was exhibited with ≤22 mm and ≤27 mm against ceftazidime and cefotaxime were observed respectively. Followed result of double disc combination and phenotypic confirmation experiments results were clearly stated that the selected *K. pneumoniae* was ESBL producer. The ceftazidime, cefotaxime and ceftazidime/clavulanic acid and cefotaxime/clavulanic acid were exhibited with merged zones and ≥5 mm zones around the combination disc when compared with disc alone were observed. All the ESBL detection test results were clearly indicated that the selected *K. pneumoniae* strain was ESBL producer.

Introduction

In many countries, multi drug resistant (MDR) bacteria and their prevalence are endemic situation and it affects acute infections with increased mortality rate [1]. Worldwide, the treating of MDR bacteria is a big challenge for clinicians due to the mismatched antibiotics consumption. In addition, the persistent of resistant may continue frequently until the detection of specific

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antibiotics or drugs [2]. Sometimes, it cause non-negligible risk of prevent the infections from active drugs or antibiotics; because of remain unfavorable condition of treating antibiotics. In this perspective, reduce the sever cause of infections, we should eradicate the pathogens in in initial stage with the help of antimicrobial sensitivity test and it can help to reduce the time to response. Also, it helps to avoid misuse of unauthorized antibiotics selection with antimicrobial stewardship principles [3,4].

Sometimes, poor selection of antibiotics may cause side effect with other adjacent infections like lymphadenopathy, urethritis, nephritis, kidney stones, kidney damage and kidney related infections [5,6]. WHO prioritized for only few antibiotics and those are available in market, even it has not potentially active [7]. Some of the antibiotics still are in last stage to come out to the clinical field. Based on the researcher identification, there is no any better antibiotic choice for treat MDR bacteria. Especially, most of the enterobacteriacea *Acinetobacter baumannii*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Proteus mirabilis*, and some others as MDRs and treatment is questionable due to the enzyme production, so it comes under in critical list of WHO [8,9]. Most of the gram negative bacteria are the dangerous pathogens and the research and development of new or effective antibiotics are urgently needed to eradicate. After evaluation of available antibiotics is very effectiveness and weaken the strength against conventional antibiotics [10]. Also, the evidences are not favor to antibiotics result, it highly favorites to pathogens due to the increased morbidity and mortality. So, based on the limited evidences, the prevent and eradication of the MDR pathogens using recommended antibiotics are very difficult. It may effect sometimes based on the antibiotic combinations [11].

Among these enterobacteriacea and their virulence to cause antimicrobial resistant is very high in *K. pneumoniae* compared to others. It is a Gram negative, rod-shaped, non-motile bacteria. It grows better in facultative anaerobic condition, also encapsulated with fermentative lactose formation. It is very most causative agent of pnemococcal infection associated with more death rates [12]. The more inflammatory response of *K. pneumoniae* associated with related human parts and cause excessive neutrophil and macrophage infiltration, excessive synthesis of cytokines. All these factors are compromised to current or available antibiotics and cause lung damage. The extended inflammation and lung injury creates pathogen dissemination in all the parts of body and developed resistant effect against all the existing antibiotics [13].

In recent years, the eradication of MDR *K. pneumoniae* by combination of multiple antibiotics is the alternative choice and also efficient in clinical department. Based on this approach, the current work is concentrated on combination of multiple antibiotics very effective against MDR *K. pneumoniae*.

Materials and methods

Collection of samples

The 100 no of multi drug resistant producing *K. pneumoniae* strains were collected from KAP Viswanathan Medical College & Hospital, Tiuchirappalli, Tamil Nadu, India. All the morphological and biochemical characterization of confirmation was confirmed by Microbiology Unit, Department of Marine Science, Bharathidasan University, Tiruchirappalli, Tamil Nadu, India. All the samples were sought to permission of ethical commeetee, Bharathidasan University, Tiruchirappalli, Tamil Nadu, India. All the chemicals, media, glass wares antimicrobial sensitivity discs were procured from Suresh Scientific & Co, Tiruchirappalli, Tamil nadu, India.

Screening of multi drug resistant K. pneumoniae

According to the Clinical & Laboratory Standards Institute (CLSI) guidelines, all the collected bacteria were initially performed against *K. pneumoniae* by disc diffusion method using specific standard antibiotic discs for detect the multi-drug resistant effect [14–17]. After carefully cultured the collected strains in nutrient broth with 24 h, the cultures were swabbed on nutrient agar plates and gently put the antibiotics on the agar surface including methicillin (5 μg), piperacillin (100 μg), amoxicillin (30 μg), cefotaxime ((10 μg), cefrixiome (30/10 μg), piperacillin/Tazobactum (100/10). All the plates were incubated at 37 °C for one day for made it zone of clearance. After, the results were interpreted based on the zone of clearance and then noted the results.

Screening of ESBL producing K. pneumoniae *strains*

After confirmation of multi drug resistant effect, particular strains were taken to detect the ESBL producing ability by disc diffusion assay using specific ESBL disc detection method [18]. In this screening test, piperacillin, ceftazidime, cefotxime, ceftazidime/clavulanic acid and cefotaxime/clavulanic acid were used. Consecutively, the ceftazidime/clavulanic acid and cefotaxime/clavulanic acid were separately tested for confirmation of ESBL production. After swab the agar surface with pathogens, all the plates were incubated with 37 °C for one day time interval. Then, the zone of inhibition was measured based on the observed plate around the discs and interpreted according to the guidelines.

Double disc combination test

All the multi-drug resistant strains were performed against selected antibiotics showed merged zone of inhibition around the discs confirmed as ESBL positive and the method was named as double disc combination method followed by previously reported evidence of Mane et al. [19]. In this experiment, the antibiotics discs of Cefotaxime/clavulanic acid combination, ceftazidime and piperacillin alone were used. After swab with 24 h old pathogenic culture on the nutrient agar surface, the discs were gently placed and stored at room temperature one day. Then, the plates were taken and seen the zones around the discs were calculated and confirmed the ESBL production.

Phenotypic positive test of ESBL production

Based on the previous report, the phenotypic confirmation tests were very effective for detection of ESBL production in tested pathogens by agar disc diffusion method followed previous reports of da Silva et al. [20]. The antibiotic discs of ceftazidime and cefotaxime were tested in the same plates, and also ceftazidime/clavulanic acid and cefotaxime/clavulanic acid in the separate plates were used for detection of ESBL in pathogens swabbed plate. Piperacillin was used as a control for negative ESBL producer. After swab, the discs were pressed on the agar surface gently and incubated at room atmosphere one day. After one day, the zones of inhibition around the performed discs were noted and interpreted the results based on the CLSI guidelines.

Result

Detection of multi drug resistant bacteria

Among the collected strains, the multi-drug resistant *K. pneumoniae* was separately screened based on the CLSI guidelines. In result, the antibiotic discs of piperacillin, cefotaxime, amoxicillin, methicillin, cefrioxime, piperacillin/tazobactum were shown no any zone

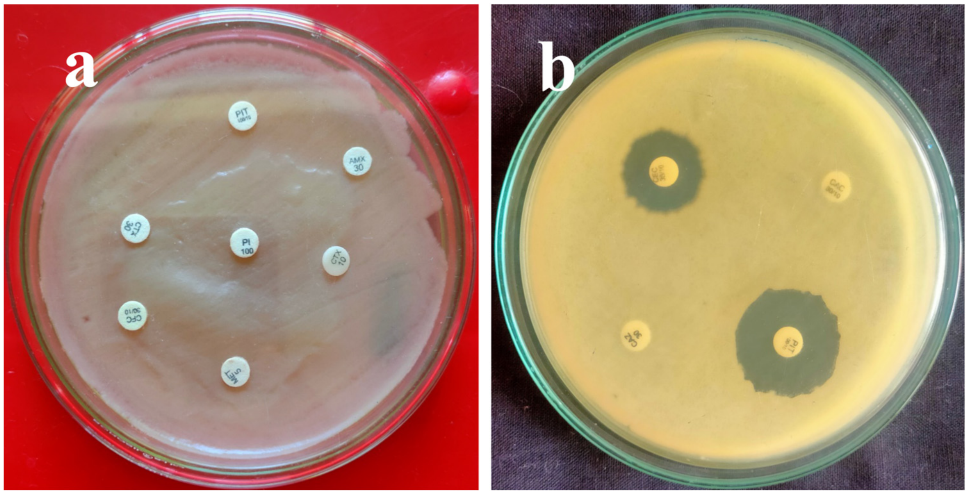

Fig. 1. Screening of multi drug resistant *K. pneumoniae* from collected strains.

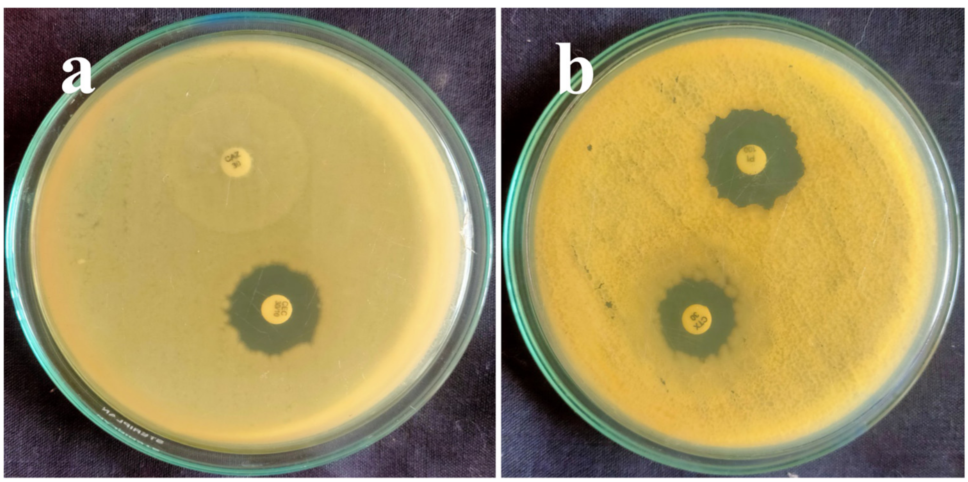

Fig. 2. Primary detection of ESBL identification test by disc diffusion assay.

of inhibition in agar well diffusion method and the result was confirmed that the selected strain was multi drug resistant strain. In addition, the discs of ceftazidime and ceftazidime/clavulanic acid also exhibited with no zone of inhibition against *K. pneumoniae* was observed. Apart from these antibiotics, the discs of cefotaxime/clavulanic acid and piperacillin/tazobactum exhibited zone of inhibition against selected *K. pneumoniae* strain. The zone range was 14 mm and 22 mm were seen respectively (Fig. 1a, b). It sought to confirm, the selected pathogen was multi drug resistant pathogen based on the CLSI guidelines. All the second generation and third generation antibiotics were highly sensitive due to the production of more virulence in the *K. pneumoniae*. The low zones of the antibitics were also indicated as sensitive due to the poor beta lactamase inhibition ability. Based on the CLSI guidelines, those antibotics zones were not reached the needed ability. So, all those low zone antibiotics were also very sensitive to selected *K. pneumoniae* [21,22]. The virulence factors of the bacteria could develop more resistant against tested antibiotics and it remains resistant characteristic nature. Piperacillin has polar chide chain and it used to help the piperacillin to enter easily into the cell wall of gram negative bacteria. After entry, it reduce the pathogenicity in bacteria and responsible gens. Finally, all the proteins and stimulating factors are inactivate and lost their beta lactamase production [15–17].

Further, the discs of piperacilli/tazobactum and ceftazidime/clavulanic acids are highly suitable for selected *K. pneumoniae*. Selected *K. pneumoniae* strains were highly more sensitive to these two piperacillin/tazobactum and ceftazime.clavulanic acid. The observed result was highly correlated to WHO statement, and *K. pneumoniae* was comes under very critical list [23]. In addition, some of the tested second and third generation cephalosporin are more sensitive to selected *K. pneumoniae*, particularly, piperacillin/tazobactum and ceftazidime/clavulanic acid were more resistant against selected *K. pneumoniae*. It brings forward to convey the message of inhibition effect against multi drug resistant bacteria. Piperacillin/tazobactum and cefotaxime are the combination drugs which used in most of the developed countries to treat antibiotic resistant pathogens. Especially, they were used for inhibit the belta lactamase producing *K. pneumoniae*. Recent report of Al-Tawfiq et al. [24], documented that the multi-drug resistant bacteria was more sensitive to piperacillin. Piperacillin was more effective against tested bacteria due to the internal damage of bacteria. Elnasser et al. [25] reported that the bacteria was produced some virulence factors in gram negative bacteria including, quorum sensing, efflux pump, ESBL production, biofilm formation and some other factors. For the reason of development of multi drug resistant effect in gram negative bacteria is production

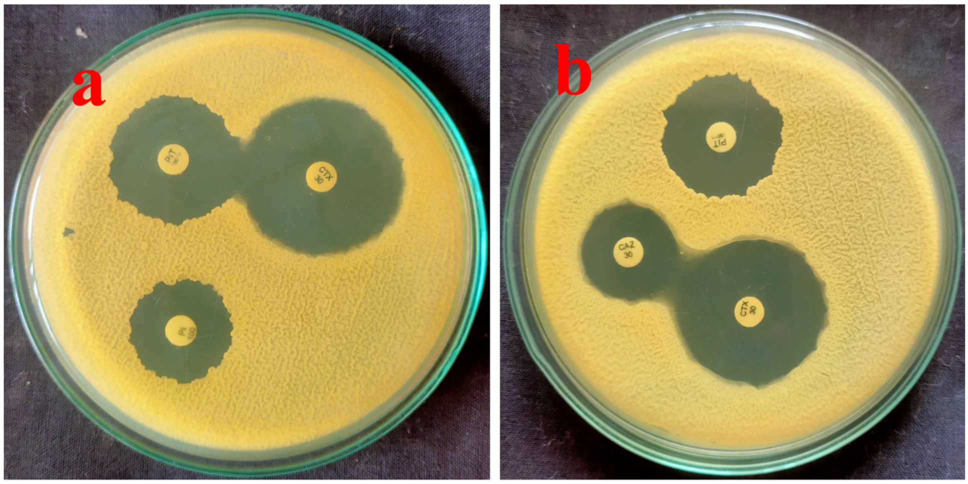

Fig. 3. Double disc combination assay of ESBL identification by disc diffusion assay.

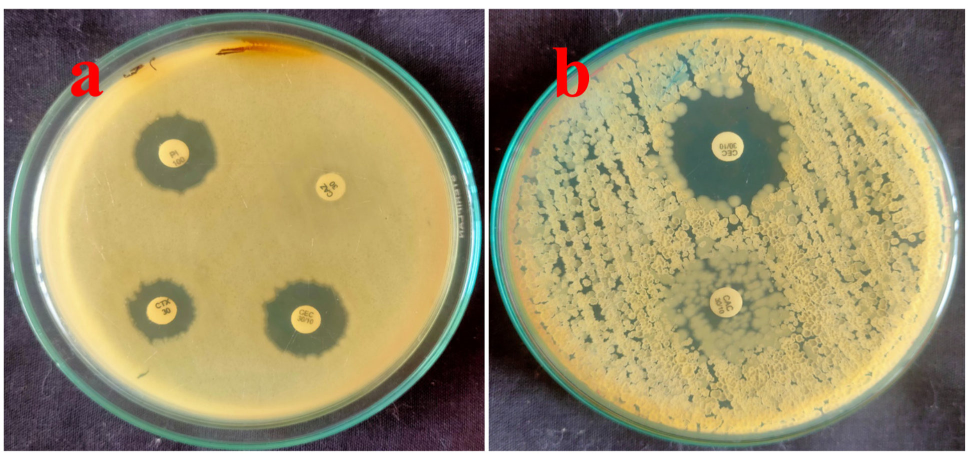

Fig. 4. Phenotypic disc diffusion assay of ESBL identification by disc diffusion assay.

of more signaling factors. All those signaling factors linked each other and produced more intracellular pathogenicity molecules [26]. The no zones around the tested discs were confirmed that the bacteria developed rigid cell wall and lipid content in their surface. Dowzicky and Chmelarov [27] also stated that the reason for development of multi drug resistant effect in gram negative bacteria is continuous usage of unauthorized medicine. Hence, the proposed study was revealed that the selected *K. pneumoniae* was more resistant behavior and zone formation of the antibiotics were confirmed by minimum inhibition concentration assays.

Initial identification of ESBL formation

The initial ESBL identification test result was more supported to *K. pneumoniae* due to the more sensitive nature against tested ESBL antibiotics. Because, all the antibiotics were lost their efficacy against tested *K. pneumoniae* due to the virulence factors activation. In total result, almost 90% of the bacteria were more sensitive to tested third generation cephalosporin antibiotics. They could not develop multi drug resistant effect and also absence of ESBL production. In this current result, the specific ESBL identification antibiotics of ceftazidime and cefotaxime were exhibited no zones and 10 mm zone of clearance against cefotaxime and cefotaxime were observed against tested *K. pneumoniae* strain. In addition, CLSI guidelines of below the zones of ≤22 mm and ≤27 mm against were observed (Fig. 2a, b). The initial ESBL identification experiment is more reliable to detect the ESBL production in gram negative bacteria [28,29]. So, our result was more correlated with present result, and confirmed that the selected *K. pneumoniae* was ESBL producer. Recent researchers were also agreed and supported evidence to the current result, the ceftazidime and cefotaxime is the better choice to detect ESBL producing bacteria. Apart from this, the respective combination discs of cefotaxime/clavualnic acid and piperacillin were showed 18 mm and 22 mm. This evidence was also more correlated with CLSI guidelines and revealed that the selected *K. pneumoniae* strain was ESBL producer.

Detection of ESBL production using double disc diffusion method

In secondary identification of ESBL detection, the discs of piperacillin, cefotaxime and piperacillin/tazobactum were shown more zones against tested *K. pneumoniae* strains. It was exhibited 14 mm, 30 mm and 24 mm zone of inhibition against tested K. pneumonia was observed for respective antibiotics (Fig. 3a, b). In addition, the cross checks of the piperacillin/tazobactum in second plate were also confirmed the same zone of 30 mm. This result was predicted the selected *K. pneumonae* strain as an ESBL positive strains. The combination disc was ≥5 mm in zone of inhibition compared with discs alone were observed clearly [21,22,30]. In total, more than 90% of the strains were more sensitive to selected antibiotics except selected *K. pneumoniae* strain. So, our result was more favorable to selected pathogen and it is highly multi drug resistant pathogen and also ESBL producer [31,32].

Phenotypic confirmation of ESBL production in selected strain

The final phenotypic confirmation experiment of ESBL identification in this current research was also more favorable to selected *K. pneumoniae* due to the better invade effect against antibiotics. The antibiotics were totally lost their effect due to cleavage of beta lactum ring. After the nature of the antibiotics was lost all the characteristic nature and *K. pneumoniae* remain withstand in the original characters. *K. pneumonia* produced more spores and other bacterial characteristics easily. So, all the current antibiotics were inability against tested *K. pneumoniae*. In this result, ceftazidime, cefotaxime, piperacillin, ceftazidime/tazobactum were shown no zone, 6 mm, 10 mm and 8 mm were observed respectively (Fig. 4a, b). It confirms that the antibiotics were not better against tested *K. pneumoniae* strain. Altogether, the current result was conveyed that the selected *K. pneumoniae* was multi drug resistant bacteria due to the ESBL production. All the antibiotics usage of phenotypic confirmation study results was more favorable to *K. pneumoniae* for ESBL production.

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Competing interests

None declared.

Ethical consideration

The samples were approved by the ethics review committee (S. No of IEC Management office:DM/2016/101/55) from the Department of Microbiology, Bharathidasan University, Tiruchirappalli, Tamil Nadu, India. The Permission was sought from the hospital and laboratory authorities. The ethical principles of scientific research as well as related national laws and regulations were adhered to.

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